

JOE reference manual

by Marco Bertacca

Package
com.veryant.joe

com.veryant.joe

Class Block

```

java.lang.Object
  |
  +-- java.util.AbstractCollection
        |
        +-- java.util.AbstractList
              |
              +-- java.util.ArrayList
                    |
                    +-- com.veryant.joe.Block
  
```

All Implemented Interfaces:

InternalObject, java.util.Collection, java.util.List, java.io.Serializable, java.lang.Cloneable, java.util.RandomAccess, java.util.List

public class **Block**
 extends java.util.ArrayList
 implements java.util.List, java.util.RandomAccess, java.lang.Cloneable, java.io.Serializable,
 java.util.List, java.util.Collection, InternalObject

This class implements a JOE block.

Method Summary

Block	\$extends (Block b)
Block	\$new () Returns a JOE Object obtained by cloning this block.
Block	\$new (java.lang.Object[] args) same as \$new() with parameters
Block	add ()
Block	add (java.lang.Object[] args)
java.lang.Object	clone () Returns a clone of this object.
java.lang.Object	doWhileFalse (Block b) Executes this block and then executes <i>b</i> : if the result of <i>b</i> execution is false then executes itself again.
java.lang.Object	doWhileTrue (Block b) Executes this block and then executes <i>b</i> : if the result of <i>b</i> execution is true then executes itself again.
java.lang.Object	exec () Executes the chains of methods of this block.
java.lang.Object	exec (java.lang.Object[] argv) Executes the chains of methods of this block with the supplied arguments.

java.lang.Object	exec (java.lang.Object[] vars, java.lang.Object[] argv)
java.lang.Object[]	getArgv () Returns an array containing all the variables values actually passed to this block as arguments.
int	getCol ()
java.lang.Object	getJoeClass ()
int	getRow ()
Variable	getSetLocalVariable (java.lang.String name)
Variable	getSetVariable (java.lang.String name)
java.lang.Object	getVariable (Variable var)
java.lang.Object	getVariable (wString name) Returns the value of the a variable whose name is <i>name</i> .
java.lang.String[]	getVariablesNames () Returns an array containing all the variables names that can be accessed by this block.
java.lang.Object	init ()
java.lang.Object	init (java.lang.Object[] argv)
Variable	lookForVariable (java.lang.String name)
Variable	lookForVariable (java.lang.String name, int[] depth)
java.lang.Object	multiply ()
java.lang.Object	multiply (java.lang.Object[] argv)
java.lang.String	name () Returns the name of the block
Block	setParent (Block b)
java.lang.Object	setVariable (Variable var, java.lang.Object val)
java.lang.Object	setVariable (wString name, java.lang.Object val) Assigns or create a variable whose name is <i>name</i> with the value <i>val</i> .
java.lang.String	toString ()
java.lang.Object	whileFalse (Block b) Executes this block and if its result is false then executes <i>b</i> .

java.lang.Object	<u>whileTrue</u> (<u>Block</u> b) Executes this block and if its result is <code>true</code> then executes <i>b</i> .
------------------	---

Methods inherited from class `java.util.ArrayList`

```
add, add, addAll, addAll, clear, clone, contains, ensureCapacity, equals, forEach,
get, hashCode, indexOf, isEmpty, iterator, lastIndexOf, listIterator, listIterator,
remove, remove, removeAll, removeIf, replaceAll, retainAll, set, size, sort,
spliterator, subList, toArray, toArray, trimToSize
```

Methods inherited from class `java.util.AbstractList`

```
add, add, addAll, clear, equals, get, hashCode, indexOf, iterator, lastIndexOf,
listIterator, listIterator, remove, set, subList
```

Methods inherited from class `java.util.AbstractCollection`

```
add, addAll, clear, contains, containsAll, isEmpty, iterator, remove, removeAll,
retainAll, size, toArray, toArray, toString
```

Methods inherited from class `java.lang.Object`

```
equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait
```

Methods inherited from interface `java.util.Collection`

```
add, addAll, clear, contains, containsAll, equals, hashCode, isEmpty, iterator,
parallelStream, remove, removeAll, removeIf, retainAll, size, spliterator, stream,
toArray, toArray, toArray
```

Methods inherited from interface `java.lang.Iterable`

```
forEach, iterator, spliterator
```

Methods inherited from interface `java.util.List`

```
add, add, addAll, addAll, clear, contains, containsAll, copyOf, equals, get,  
hashCode, indexOf, isEmpty, iterator, lastIndexOf, listIterator, listIterator, of,  
of, of, of, of, of, of, of, of, of, of, of, of, remove, remove, removeAll, replaceAll,  
retainAll, set, size, sort, spliterator, subList, toArray, toArray
```

Methods inherited from interface `java.util.Collection`

```
add, addAll, clear, contains, containsAll, equals, hashCode, isEmpty, iterator,
parallelStream, remove, removeAll, removeIf, retainAll, size, spliterator, stream,
toArray, toArray, toArray
```

Methods inherited from interface `java.lang.Iterable`

```
forEach, iterator, spliterator
```

Methods inherited from interface `java.util.List`

```
add, add, addAll, addAll, clear, contains, containsAll, copyOf, equals, get,  
hashCode, indexOf, isEmpty, iterator, lastIndexOf, listIterator, listIterator, of,  
of, of, of, of, of, of, of, of, of, of, of, of, remove, remove, removeAll, replaceAll,  
retainAll, set, size, sort, spliterator, subList, toArray, toArray
```

Methods inherited from interface java.util.Collection

add, addAll, clear, contains, containsAll, equals, hashCode, isEmpty, iterator, parallelStream, remove, removeAll, removeIf, retainAll, size, spliterator, stream, toArray, toArray, toArray

Methods inherited from interface java.lang.Iterable

forEach, iterator, spliterator

Methods

exec

```
public final java.lang.Object exec()
    throws JOEException
```

Executes the chains of methods of this block.

multiply

```
public final java.lang.Object multiply()
    throws JOEException
```

See Also:

[exec\(\)](#)

exec

```
public final java.lang.Object exec(java.lang.Object[] argv)
    throws JOEException
```

Executes the chains of methods of this block with the supplied arguments.

multiply

```
public final java.lang.Object multiply(java.lang.Object[] argv)
    throws JOEException
```

See Also:

[\(Object...argv\).](#)

init

```
public final java.lang.Object init()
    throws JOEException
```

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init

```
public final java.lang.Object init(java.lang.Object[] argv)
    throws JOEException
```

exec

```
public final java.lang.Object exec(java.lang.Object[] vars,
    java.lang.Object[] argv)
    throws JOEException
```

whileTrue

```
public final java.lang.Object whileTrue(Block b)
    throws JOEException
```

Executes this block and if its result is `true` then executes *b*. Repeats until this block returns an object not equal to `true`. Returns the result of *b* last execution.

whileFalse

```
public final java.lang.Object whileFalse(Block b)
    throws JOEException
```

Executes this block and if its result is `false` then executes *b*. Repeats until this block returns an object not equal to `false`. Returns the result of *b* last execution.

doWhileTrue

```
public final java.lang.Object doWhileTrue(Block b)
    throws JOEException
```

Executes this block and then executes *b*; if the result of *b* execution is `true` then executes itself again. Repeats until *b* returns an object not equal to `true`

doWhileFalse

```
public final java.lang.Object doWhileFalse(Block b)
    throws JOEException
```

Executes this block and then executes *b*; if the result of *b* execution is `false` then executes itself again. Repeats until *b* returns an object not equal to `false`

setVariable

```
public java.lang.Object setVariable(WString name,
    java.lang.Object val)
    throws JOEException
```

Assigns or create a variable whose name is *name* with the value *val*.

setVariable

```
public java.lang.Object setVariable(Variable var,
    java.lang.Object val)
```

getVariable

```
public java.lang.Object getVariable(WString name)
```

Returns the value of the a variable whose name is *name*.

getVariable

```
public java.lang.Object getVariable(Variable var)
```

getVariablesNames

```
public java.lang.String[] getVariablesNames()
```

Returns an array containing all the variables names that can be accessed by this block.

getArgv

```
public java.lang.Object[] getArgv()
```

Returns an array containing all the variables values actually passed to this block as arguments.

clone

```
public java.lang.Object clone()
```

Returns a clone of this object.

\$new

```
public Block $new()  
    throws JOEException
```

Returns a JOE Object obtained by cloning this block. The block is executed and every variable assigned to a block will be treated as a method of the JOE Object object. This method can be invoked on a JOE Object too.

add

```
public Block add()  
    throws JOEException
```

See Also:

[\\$new\(\)](#)

\$new

```
public Block $new(java.lang.Object[] args)  
    throws JOEException
```

same as \$new() with parameters

See Also:

[\\$new\(\)](#)

add

```
public Block add(java.lang.Object[] args)
    throws JOEException
```

See Also:

[\(Object...args\)](#)

name

```
public java.lang.String name()
```

Returns the name of the block

getRow

```
public int getRow()
```

getCol

```
public int getCol()
```

setParent

```
public Block setParent(Block b)
```

\$extends

```
public Block $extends(Block b)
```

getJoeClass

```
public java.lang.Object getJoeClass()
```

toString

```
public java.lang.String toString()
```

lookForVariable

```
public Variable lookForVariable(java.lang.String name,
    int[] depth)
```

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lookForVariable

```
public Variable lookForVariable(java.lang.String name)
```

getSetLocalVariable

```
public Variable getSetLocalVariable(java.lang.String name)
```

getSetVariable

```
public Variable getSetVariable(java.lang.String name)
```

com.veryant.joe Class CommandBase

java.lang.Object

└-com.veryant.joe.CommandBase

Direct Known Subclasses:

[DefaultCommand](#)

```
public class CommandBase
extends java.lang.Object
```

This class contains some methods that are useful for any flavor of JOE implementation.

Constructor Summary

public	CommandBase()
--------	-------------------------------

Method Summary

void	\$extends (Block child, Block parent) Makes the child inheriting from parent.
boolean	\$instanceof (java.lang.Object obj, java.lang.Class clazz) Similar to the java operator instanceof.
boolean	\$instanceof (java.lang.Object obj, java.lang.String clazz) Convenience method for instanceof (obj, getClass(clazz)).
java.lang.Object	\$new (java.lang.String fname) Executes the specified JOE script and returns it as an object.
java.lang.Object	\$new (java.lang.String fname, java.lang.Object[] argv) Executes the specified JOE script and returns it as an object.
void	\$throw (java.lang.String msg) Throws an Exception whose message is the argument.
void	\$throw (java.lang.Throwable ex) Throws a throwable passed as argument.
java.lang.Object	addPath (java.lang.String fname) Adds one path from which to load new object.
boolean	areSameObject (java.lang.Object o1, java.lang.Object o2) Returns true if the passed objects are the same object, false otherwise.
java.lang.Object[]	array (java.lang.Object[] items) Returns an Object array whose items are the parameters
int	asc (java.lang.String a) Returns a char representing the codepoint corresponding to the first letter in a string.

java.lang.String	<code>chr</code> (java.lang.Integer n) Returns a string one character long containing the character whose codepoint is specified as argument.
java.lang.String	<code>debug</code> () Turns on the debugger.
java.lang.Class	<code>getClass</code> (java.lang.String clazz) Returns the Class instance of clazz.
ClassReference	<code>getClassRef</code> (java.lang.String clazz) Returns the ClassReference instance of clazz.
java.lang.String	<code>getCwd</code> () Returns the current directory.
java.lang.Object	<code>getField</code> (java.lang.Object obj, java.lang.String name) Returns a field whose name is name of the object obj.
java.lang.String[]	<code>getPath</code> () Returns the path set.
java.lang.Object	<code>getStaticField</code> (java.lang.String clazz, java.lang.String name) Returns a static field whose name is name of the class clazz.
java.net.URL	<code>getURL</code> (java.lang.String spec) Returns an URL based on the base URL of the first script run.
boolean	<code>isNull</code> (java.lang.Object obj) Returns true if the passed object is null, false otherwise.
boolean	<code>isThrowable</code> (java.lang.Object obj) Returns the result of obj instanceof Throwable
int	<code>joe</code> (java.lang.Object[] argv) Executes the specified JOE script and returns its return code.
java.lang.Object	<code>newArray</code> (java.lang.Class clazz, int n) Returns an Object array of the specified class whose length is n.
java.lang.Object[]	<code>newArray</code> (int n) Returns an Object array whose length is n.
java.lang.Object	<code>newArray</code> (java.lang.String clazz, int n) Returns an Object array of the specified class whose length is n.
java.lang.Object	<code>newInstance</code> (java.lang.Class clazz) Returns a new instance of the specified class.
java.lang.Object	<code>newInstance</code> (java.lang.Class clazz, java.lang.Object[] args) Returns a new instance of the specified class passing the arguments args to its constructor.
java.lang.Object	<code>newInstance</code> (java.lang.String className) This is a convenience method for newInstance (getClass (className)).
java.lang.Object	<code>newInstance</code> (java.lang.String clsName, java.lang.Object[] args) This is a convenience method for newInstance (getClass (clsName), args).

java.lang.Object	newInterface (java.lang.String intf, Block blk) Returns the specified interface whose method must be implemented in the supplied block in a way similar to java.lang.reflect.Proxy.
java.lang.String	nl () Returns a string containing the line terminator.
void	systemExit () Invoke System.exit (0).
void	systemExit (int ec) Invoke System.exit (ec).
java.lang.String	toString ()
java.lang.String	typename (java.lang.Object obj) Returns the JOE name of the object.
java.lang.String	version () Returns the version.

Methods inherited from class java.lang.Object

equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructors

CommandBase

```
public CommandBase()
```

Methods

newInstance

```
public java.lang.Object newInstance(java.lang.Class clazz)
    throws java.lang.Exception
```

Returns a new instance of the specified class.

newInstance

```
public java.lang.Object newInstance(java.lang.String className)
    throws java.lang.Exception
```

This is a convenience method for newInstance (getClass (className)).

newInstance

```
public java.lang.Object newInstance(java.lang.Class clazz,
    java.lang.Object[] args)
    throws java.lang.Exception
```

Returns a new instance of the specified class passing the arguments args to its constructor.

newInstance

```
public java.lang.Object newInstance(java.lang.String clsName,  
                                     java.lang.Object[] args)  
    throws java.lang.Exception
```

This is a convenience method for `newInstance (getClass (clsName), args)`.

newArray

```
public java.lang.Object[] newArray(int n)  
    throws java.lang.Exception
```

Returns an Object array whose length is n.

newArray

```
public java.lang.Object newArray(java.lang.Class clazz,  
                                   int n)  
    throws java.lang.Exception
```

Returns an Object array of the specified class whose length is n.

newArray

```
public java.lang.Object newArray(java.lang.String clazz,  
                                   int n)  
    throws java.lang.Exception
```

Returns an Object array of the specified class whose length is n.

array

```
public java.lang.Object[] array(java.lang.Object[] items)  
    throws java.lang.Exception
```

Returns an Object array whose items are the parameters

getField

```
public java.lang.Object getField(java.lang.Object obj,  
                                 java.lang.String name)  
    throws java.lang.Exception
```

Returns a field whose name is name of the object obj.

getStaticField

```
public java.lang.Object getStaticField(java.lang.String clazz,  
                                         java.lang.String name)  
    throws java.lang.Exception
```

Returns a static field whose name is name of the class clazz.

getClass

```
public java.lang.Class getClass(java.lang.String clazz)  
    throws java.lang.Exception
```

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Returns the Class instance of clazz.

getClassRef

```
public ClassReference getClassRef(java.lang.String clazz)
    throws java.lang.Exception
```

Returns the ClassReference instance of clazz. The ClassReference object can be used to invoke a static method of a class, e.g. the ClassReference of java.lang.System can be used to invoke System.getProperty(String prop).

newInterface

```
public java.lang.Object newInterface(java.lang.String intf,
    Block blk)
    throws java.lang.Exception
```

Returns the specified interface whose method must be implemented in the supplied block in a way similar to java.lang.reflect.Proxy.

systemExit

```
public void systemExit(int ec)
```

Invoke System.exit (ec).

systemExit

```
public void systemExit()
```

Invoke System.exit (0).

\$throw

```
public void $throw(java.lang.String msg)
    throws java.lang.Exception
```

Throws an Exception whose message is the argument.

\$throw

```
public void $throw(java.lang.Throwable ex)
    throws java.lang.Throwable
```

Throws a throwable passed as argument.

isNull

```
public boolean isNull(java.lang.Object obj)
```

Returns true if the passed object is null, false otherwise.

areSameObject

```
public boolean areSameObject(java.lang.Object o1,
    java.lang.Object o2)
```

Returns true if the passed objects are the same object, false otherwise.

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isThrowable

```
public boolean isThrowable(java.lang.Object obj)
```

Returns the result of obj instanceof Throwable

\$instanceof

```
public boolean $instanceof(java.lang.Object obj,  
    java.lang.Class clazz)
```

Similar to the java operator instanceof.

typename

```
public java.lang.String typename(java.lang.Object obj)
```

Returns the JOE name of the object.

addPath

```
public java.lang.Object addPath(java.lang.String fname)  
    throws java.lang.Exception
```

Adds one path from which to load new object.

getPath

```
public java.lang.String[] getPath()
```

Returns the path set.

\$new

```
public java.lang.Object $new(java.lang.String fname)  
    throws java.lang.Exception
```

Executes the specified JOE script and returns it as an object.

\$new

```
public java.lang.Object $new(java.lang.String fname,  
    java.lang.Object[] argv)  
    throws java.lang.Exception
```

Executes the specified JOE script and returns it as an object.

joe

```
public int joe(java.lang.Object[] argv)  
    throws java.lang.Exception
```

Executes the specified JOE script and returns its return code.

\$instanceof

```
public boolean $instanceof(java.lang.Object obj,  
    java.lang.String clazz)
```

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Convenience method for instanceof (obj, getClass(clazz)).

nl

```
public java.lang.String nl()
```

Returns a string containing the line terminator.

chr

```
public java.lang.String chr(java.lang.Integer n)
```

Returns a string one character long containing the character whose codepoint is specified as argument.

asc

```
public int asc(java.lang.String a)
```

Returns a char representing the codepoint corresponding to the first letter in a string.

\$extends

```
public void $extends(Block child,  
                    Block parent)
```

Makes the child inheriting from parent.

getURL

```
public java.net.URL getURL(java.lang.String spec)  
    throws java.net.MalformedURLException
```

Returns an URL based on the base URL of the first script run.

version

```
public java.lang.String version()
```

Returns the version.

getcwd

```
public java.lang.String getcwd()
```

Returns the current directory.

debug

```
public java.lang.String debug()  
    throws JOEException
```

Turns on the debugger.

toString

```
public java.lang.String toString()
```

com.veryant.joe Class DefaultCommand

java.lang.Object

└ [com.veryant.joe.CommandBase](#)

└ **com.veryant.joe.DefaultCommand**

public class **DefaultCommand**
extends [CommandBase](#)

This class contains methods of ! standard.

Nested Class Summary

class	DefaultCommand.Switch DefaultCommand.Switch
-------	--

Constructor Summary

public	DefaultCommand()
--------	----------------------------------

Method Summary

java.lang.Object	\$break() Causes the exit from the current block.
java.lang.Object	\$break(java.lang.String blkName) Causes the exit from the block whose name is blkName.
java.lang.Object	\$for(int start, int end, Block code) This is a convenience method for for (start, end, 1).
java.lang.Object	\$for(int start, int end, int step, Block code) This method implements a 'for' cycle.
java.lang.Object	\$if(Block cond, Block ifTrue) if implementation without 'else'.
java.lang.Object	\$if(Block cond, Block ifTrue, Block ifFalse) if implementation with 'else'.
java.lang.Object	\$if(java.lang.Boolean cond, Block ifTrue) If cond is true then the block ifTrue is executed and its return value returned, otherwise cond is returned.
java.lang.Object	\$if(boolean cond, Block ifTrue, Block ifFalse) If cond is true then the block ifTrue is executed and its return value returned, otherwise ifFalse is executed and its return value returned.
java.lang.Object	\$switch(java.lang.Object cfrt) Returns an instance of class Switch created using cfrt as parameter.

java.lang.Object	\$try (Block blk) Executes the block passed as argument: if the execution throws an exception then the exception is returned instead of the expected return object.
java.lang.Object	\$try (Block blk, Block excpt) Executes the block passed as 1st argument: if the execution throws an exception then the block passed as 2nd argument is executed with the exception as argument.
java.lang.Object	\$while (Block cond, Block code) This method implements a loop.
java.lang.Object	arraySort (java.lang.Object[] array, Block blk) Sort an array according to the block.
java.lang.Object	binarySearch (java.lang.Object[] array, java.lang.Object key, Block blk) Binary search on an array according to the block.
void	breakLoop () Causes the exit from a loop.
java.lang.Object	doWhile (Block code, Block cond) This method implements a loop.
java.lang.Object	eprint (java.lang.Object[] b) Prints the specified objects on stderr.
java.lang.Object	eprintln () Prints a newline on stderr.
java.lang.Object	eprintln (java.lang.Object[] b) Prints the specified objects followed by a newline on stderr.
int	exec (java.lang.Object[] cmds) Executes the specified command and returns its return code.
int	exec (java.lang.ProcessBuilder pb) Executes the specified ProcessBuilder
int	execFromDir (java.io.File dir, java.lang.Object[] cmds) Executes the specified command from the specified directory and returns its return code.
int	execFromDir (java.lang.String dir, java.lang.Object[] cmds) Executes the specified command from the specified directory and returns its return code.
java.lang.String	execGetOut (java.lang.Object[] cmds) Executes the specified command and returns its standard output as a string.
int	execJoe (java.lang.Object[] cmds) Stops the execution of the current script and executes the script specified as argument.
java.lang.Object	foreach (byte[] list, Block code)
java.lang.Object	foreach (byte[] list, int n, Block code)
java.lang.Object	foreach (char[] list, Block code)
java.lang.Object	foreach (char[] list, int n, Block code)

java.lang.Object	foreach (java.util.Collection list, Block code) This is a convenience method for foreach (list, 0, code).
java.lang.Object	foreach (java.util.Collection list, int n, Block code) This method implements a 'for each'.
java.lang.Object	foreach (java.util.Enumeration it, Block code) This is a convenience method for foreach (list, 0, code).
java.lang.Object	foreach (java.util.Enumeration it, int n, Block code) This method implements a 'for each'.
java.lang.Object	foreach (java.lang.Iterable it, Block code) This is a convenience method for foreach (list, 0, code).
java.lang.Object	foreach (java.lang.Iterable it, int n, Block code) This method implements a 'for each'.
java.lang.Object	foreach (java.util.Iterator it, Block code) This is a convenience method for foreach (list, 0, code).
java.lang.Object	foreach (java.util.Iterator it, int n, Block code) This method implements a 'for each'.
java.lang.Object	foreach (java.lang.Object[] list, Block code) This is a convenience method for foreach (list, 0, code).
java.lang.Object	foreach (java.lang.Object[] list, int n, Block code) This method implements a 'for each'.
Glob	getGlob (java.lang.String glob, boolean caseInsensitive) Returns a Glob object that performs match operations on paths thru the method 'matches'.
boolean	isConsole ()
Pipe	pipe () Returns an object of type Pipe that allows the execution of programs in a pipeline.
java.lang.Object	print (java.lang.Object[] b) Prints the specified objects.
java.lang.Object	println () Prints a newline.
java.lang.Object	println (java.lang.Object[] b) Prints the specified objects followed by a newline.
java.lang.Object	random () Returns a random number in the range 0.0 - 1.0.
java.lang.String	readLine () Read a line from the console.
java.lang.Object	runAsBlock (Block blk, java.lang.String name, java.lang.Object[] cmds) Executes the specified JOE script as it were an inner block of the given argument.
int	runJoe (java.lang.Object[] cmds) Executes the specified JOE script and returns its return code.

java.lang.String	showInputDialog (java.lang.Object[] message) Shows the specified objects in a graphical window and reads an input that returns.
void	showMessageDialog (java.lang.Object[] message) edit Shows the specified objects in a graphical window.
void	sleep (int millis) Suspends execution for the specified number of milliseconds
java.lang.Object	switchTrue () Returns an instance of class Switch created using Boolean.TRUE as parameter.
int	system (java.lang.String[] cmd)
java.lang.String	systemGetenv (java.lang.String name) Reads the value of the specified environment variable.

Methods inherited from class [com.veryant.joe.CommandBase](#)

[\\$extends](#), [\\$instanceof](#), [\\$instanceof](#), [\\$new](#), [\\$new](#), [\\$throw](#), [\\$throw](#), [addPath](#), [areSameObject](#), [array](#), [asc](#), [chr](#), [debug](#), [getClass](#), [getClassRef](#), [getcwd](#), [getField](#), [getPath](#), [getStaticField](#), [getURL](#), [isNull](#), [isThrowable](#), [joe](#), [newArray](#), [newArray](#), [newArray](#), [newInstance](#), [newInstance](#), [newInstance](#), [newInstance](#), [newInterface](#), [nl](#), [systemExit](#), [systemExit](#), [toString](#), [typename](#), [version](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Constructors

DefaultCommand

```
public DefaultCommand()
```

Methods

isConsole

```
public boolean isConsole()
```

readLine

```
public java.lang.String readLine()  
throws java.io.IOException
```

Read a line from the console.

println

```
public java.lang.Object println()
```

(continued from last page)

Prints a newline.

eprintln

```
public java.lang.Object eprintln()
```

Prints a newline on stderr.

print

```
public java.lang.Object print(java.lang.Object[] b)
```

Prints the specified objects.

eprint

```
public java.lang.Object eprint(java.lang.Object[] b)
```

Prints the specified objects on stderr.

println

```
public java.lang.Object println(java.lang.Object[] b)
```

Prints the specified objects followed by a newline.

eprintln

```
public java.lang.Object eprintln(java.lang.Object[] b)
```

Prints the specified objects followed by a newline on stderr.

\$if

```
public java.lang.Object $if(java.lang.Boolean cond,  
    Block ifTrue)  
    throws java.lang.Exception
```

If cond is true then the block ifTrue is executed and its return value returned, otherwise cond is returned.

\$if

```
public java.lang.Object $if(boolean cond,  
    Block ifTrue,  
    Block ifFalse)  
    throws java.lang.Exception
```

If cond is true then the block ifTrue is executed and its return value returned, otherwise ifFalse is executed and its return value returned.

\$if

```
public java.lang.Object $if(Block cond,  
    Block ifTrue)  
    throws java.lang.Exception
```

if implementation without 'else'.

(continued from last page)

\$if

```
public java.lang.Object $if(Block cond,  
    Block ifTrue,  
    Block ifFalse)  
    throws java.lang.Exception
```

if implementation with 'else'.

\$while

```
public java.lang.Object $while(Block cond,  
    Block code)  
    throws JOEException
```

This method implements a loop. The code block is executed while the cond block is executed and it returns a boolean value true or until the breakLoop () invocation is encountered.

doWhile

```
public java.lang.Object dowhile(Block code,  
    Block cond)  
    throws JOEException
```

This method implements a loop. This loop will execute the code block once, before checking if the cond block returns the boolean value true, then it will repeat the loop as long as the condition is true or until the breakLoop () invocation is encountered..

\$for

```
public java.lang.Object $for(int start,  
    int end,  
    int step,  
    Block code)  
    throws JOEException
```

This method implements a 'for' cycle. The code block is executed for passing the index as argument.

\$for

```
public java.lang.Object $for(int start,  
    int end,  
    Block code)  
    throws JOEException
```

This is a convenience method for for (start, end, 1).

foreach

```
public java.lang.Object foreach(java.lang.Object[] list,  
    int n,  
    Block code)  
    throws JOEException
```

This method implements a 'for each'. The code block is executed for each object passed in the array startint from the n-th, each object is passed to the block as argument.

foreach

```
public java.lang.Object foreach(java.lang.Object[] list,  
    Block code)  
    throws JOEException
```

(continued from last page)

This is a convenience method for foreach (list, 0, code).

foreach

```
public java.lang.Object foreach(java.util.Collection list,  
    int n,  
    Block code)  
throws JOEException
```

This method implements a 'for each'. The code block is executed for each object passed in the Collection starting from the n-th, each object is passed to the block as argument.

foreach

```
public java.lang.Object foreach(java.util.Collection list,  
    Block code)  
throws JOEException
```

This is a convenience method for foreach (list, 0, code).

foreach

```
public java.lang.Object foreach(java.util.Iterator it,  
    int n,  
    Block code)  
throws JOEException
```

This method implements a 'for each'. The code block is executed for each object passed in the Iterator starting from n, each object is passed to the block as argument.

foreach

```
public java.lang.Object foreach(java.util.Iterator it,  
    Block code)  
throws JOEException
```

This is a convenience method for foreach (list, 0, code).

foreach

```
public java.lang.Object foreach(java.util.Enumeration it,  
    int n,  
    Block code)  
throws JOEException
```

This method implements a 'for each'. The code block is executed for each object passed in the Enumeration starting from n, each object is passed to the block as argument.

foreach

```
public java.lang.Object foreach(java.util.Enumeration it,  
    Block code)  
throws JOEException
```

This is a convenience method for foreach (list, 0, code).

(continued from last page)

foreach

```
public java.lang.Object foreach(java.lang.Iterable it,  
    int n,  
    Block code)  
    throws JOEException
```

This method implements a 'for each'. The code block is executed for each object passed in the Iterable.iterator starting from n, each object is passed to the block as argument.

foreach

```
public java.lang.Object foreach(java.lang.Iterable it,  
    Block code)  
    throws JOEException
```

This is a convenience method for foreach (list, 0, code).

foreach

```
public java.lang.Object foreach(byte[] list,  
    int n,  
    Block code)  
    throws JOEException
```

foreach

```
public java.lang.Object foreach(byte[] list,  
    Block code)  
    throws JOEException
```

foreach

```
public java.lang.Object foreach(char[] list,  
    int n,  
    Block code)  
    throws JOEException
```

foreach

```
public java.lang.Object foreach(char[] list,  
    Block code)  
    throws JOEException
```

breakLoop

```
public void breakLoop()  
    throws BreakLoopException
```

Causes the exit from a loop.

(continued from last page)

\$switch

```
public java.lang.Object $switch(java.lang.Object cfrt)
```

Returns an instance of class Switch created using cfrt as parameter.

switchTrue

```
public java.lang.Object switchTrue()
```

Returns an instance of class Switch created using Boolean.TRUE as parameter.

\$break

```
public java.lang.Object $break()  
    throws java.lang.Exception
```

Causes the exit from the current block.

\$break

```
public java.lang.Object $break(java.lang.String blkName)  
    throws java.lang.Exception
```

Causes the exit from the block whose name is blkName.

\$try

```
public java.lang.Object $try(Block blk)  
    throws java.lang.Exception
```

Executes the block passed as argument: if the execution throws an exception then the exception is returned instead of the expected return object.

\$try

```
public java.lang.Object $try(Block blk,  
    Block excpt)  
    throws java.lang.Exception
```

Executes the block passed as 1st argument: if the execution throws an exception then the block passed as 2nd argument is executed with the exception as argument.

exec

```
public int exec(java.lang.ProcessBuilder pb)  
    throws java.lang.Exception
```

Executes the specified ProcessBuilder

exec

```
public int exec(java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified command and returns its return code.

(continued from last page)

execFromDir

```
public int execFromDir(java.io.File dir,  
    java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified command from the specified directory and returns its return code.

execFromDir

```
public int execFromDir(java.lang.String dir,  
    java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified command from the specified directory and returns its return code.

system

```
public int system(java.lang.String[] cmd)  
    throws java.lang.Exception
```

execGetOut

```
public java.lang.String execGetOut(java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified command and returns its standard output as a string.

pipe

```
public Pipe pipe()
```

Returns an object of type Pipe that allows the execution of programs in a pipeline.

runJoe

```
public int runJoe(java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified JOE script and returns its return code.

runAsBlock

```
public java.lang.Object runAsBlock(Block blk,  
    java.lang.String name,  
    java.lang.Object[] cmds)  
    throws java.lang.Exception
```

Executes the specified JOE script as it were an inner block of the given argument.

execJoe

```
public int execJoe(java.lang.Object[] cmds)  
    throws ExecException
```

Stops the execution of the current script and executes the script specified as argument.

(continued from last page)

systemGetenv

```
public java.lang.String systemGetenv(java.lang.String name)
```

Reads the value of the specified environment variable.

arraySort

```
public java.lang.Object arraySort(java.lang.Object[] array,  
    Block blk)
```

Sort an array according to the block.

binarySearch

```
public java.lang.Object binarySearch(java.lang.Object[] array,  
    java.lang.Object key,  
    Block blk)
```

Binary search on an array according to the block.

random

```
public java.lang.Object random()
```

Returns a random number in the range 0.0 - 1.0.

sleep

```
public void sleep(int millis)  
    throws java.lang.Exception
```

Suspends execution for the specified number of milliseconds

getGlob

```
public Glob getGlob(java.lang.String glob,  
    boolean caseInsensitive)
```

Returns a Glob object that performs match operations on paths thru the method 'matches'.

showMessageDialog

```
public void showMessageDialog(java.lang.Object[] message)
```

edit Shows the specified objects in a graphical window.

showInputDialog

```
public java.lang.String showInputDialog(java.lang.Object[] message)
```

Shows the specified objects in a graphical window and reads an input that returns.

com.veryant.joe Class DefaultCommand.Switch

java.lang.Object

↓ -com.veryant.joe.DefaultCommand.Switch

public static class **DefaultCommand.Switch**
extends java.lang.Object

This class is used in order to implement a behaviour similar to the Java switch.

Method Summary

java.lang.Object	<u>\$case</u> (<u>Block</u> blk2)
java.lang.Object	<u>\$case</u> (<u>Block</u> blk2, <u>Block</u> block) This method is similar to \$case (Object cfrt2, Block block) The <i>block</i> is supposed to returns a boolean and it is executed only if no block in the switch has been executed yet.
java.lang.Object	<u>\$case</u> (java.lang.Object cfrt2) This method compares cfrt2 with the object specified in the constructor and if they are equal then the following \$case (cfrt2, block), if any, will be executed independently from its argument.
java.lang.Object	<u>\$case</u> (java.lang.Object cfrt2, <u>Block</u> block) This method executes block under the following conditions: - any previous invocation of this method on this object hasn't been executed yet; - the argument of any previous \$case(cfrt2) invocation, or the argument of this invocation is equal to the object specified in the constructor.
java.lang.Object	<u>\$default</u> (<u>Block</u> block) This method executes block if any previous invocation of \$case (cfrt2, block) method on this object hasn't be executed yet.
java.lang.Object	<u>endSwitch</u> () This method returns the return code of the block executed or null if no block has been executed by this object.

Methods inherited from class java.lang.Object

equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Methods

\$case

public java.lang.Object **\$case**(java.lang.Object cfrt2)

This method compares cfrt2 with the object specified in the constructor and if they are equal then the following \$case (cfrt2, block), if any, will be executed independently from its argument. It returns this.

(continued from last page)

\$case

```
public java.lang.Object $case(Block blk2)
    throws JOEException
```

\$case

```
public java.lang.Object $case(java.lang.Object cfrt2,
    Block block)
    throws JOEException
```

This method executes block under the following conditions: - any previous invocation of this method on this object hasn't been executed yet; - the argument of any previous `$case(cfrt2)` invocation, or the argument of this invocation is equal to the object specified in the constructor.

\$case

```
public java.lang.Object $case(Block blk2,
    Block block)
    throws JOEException
```

This method is similar to `$case (Object cfrt2, Block block)` The *block* is supposed to returns a boolean and it is executed only if no block in the switch has been executed yet.

\$default

```
public java.lang.Object $default(Block block)
    throws java.lang.Exception
```

This method executes block if any previous invocation of `$case (cfrt2, block)` method on this object hasn't be executed yet.

endSwitch

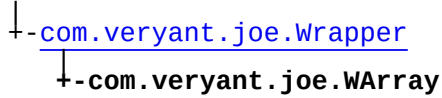
```
public java.lang.Object endSwitch()
```

This method returns the return code of the block executed or null if no block has been executed by this object.

com.veryant.joe

Class WArray

java.lang.Object



All Implemented Interfaces:
InternalObject

public class **WArray**
extends [Wrapper](#)

This class implements a JOE array and it is a wrapper of a Java array. In order to create an array from JOE you can use the following instructions:

`! array obj0 [, obj1 ... , objn ]`

which creates an array containing all the objects specified or:

`! newArray aInteger`

which creates an array of *aInteger* elements all containing null.

Constructor Summary

public	WArray (java.lang.Object[] array)
public	WArray (WNumber len)
public	WArray (int len)

Method Summary

java.lang.Object	add (java.lang.Object obj) Returns a new array longer by one of this array with <i>obj</i> in the last position.
java.lang.Object	clone () Returns a copy of this array
java.lang.Object	get (WNumber idx) Returns the object at position <i>idx</i> .
java.lang.Object	getWrapped () This is for internal use only
WInteger	length () Returns the length of this array.
java.lang.Object	set (int idx, java.lang.Object obj) Sets the object <i>obj</i> at position <i>idx</i> .
java.lang.Object	set (WNumber idx, java.lang.Object obj)
java.lang.Object	shift () Returns a new array shorter by one of this array without the object in the first position.

java.lang.Object	shift (int shft) Returns a new array shorter by <i>shft</i> of this array without <i>shft</i> objects at the beginning.
java.lang.Object	shift (WNumber shiftValue)
WInteger	size () Returns the length of this array.
java.lang.Object	slice (WNumber from, WNumber to) Returns the specified range of this array as a new array.
Wrapper.Type	type ()
java.lang.Object	unshift (java.lang.Object obj) Returns a new array longer by one of this array with <i>obj</i> in the first position.

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructors

WArray

public **WArray**(java.lang.Object[] array)

WArray

public **WArray**([WNumber](#) len)

WArray

public **WArray**(int len)

Methods

type

public Wrapper.Type **type**()

(continued from last page)

getWrapped

```
public java.lang.Object getWrapped()
```

This is for internal use only

length

```
public WInteger length()
```

Returns the length of this array.

size

```
public WInteger size()
```

Returns the length of this array.

See Also:

[length\(\)](#)

get

```
public java.lang.Object get(WNumber idx)
```

Returns the object at position *idx*.

set

```
public java.lang.Object set(int idx,  
    java.lang.Object obj)
```

Sets the object *obj* at position *idx*.

set

```
public java.lang.Object set(WNumber idx,  
    java.lang.Object obj)
```

See Also:

[set\(int, Object\)](#)

add

```
public java.lang.Object add(java.lang.Object obj)
```

Returns a new array longer by one of this array with *obj* in the last position.

shift

```
public java.lang.Object shift(int shft)
```

Returns a new array shorter by *shft* of this array without *shft* objects at the beginning.

(continued from last page)

shift

```
public java.lang.Object shift(WNumber shiftValue)
```

See Also:

[shift\(int\)](#)

shift

```
public java.lang.Object shift()
```

Returns a new array shorter by one of this array without the object in the first position.

unshift

```
public java.lang.Object unshift(java.lang.Object obj)
```

Returns a new array longer by one of this array with *obj* in the first position.

slice

```
public java.lang.Object slice(WNumber from,  
                               WNumber to)
```

Returns the specified range of this array as a new array. The value at *from* is placed at the first position in the new array. The length of the returned array will be *to* - *from*.

clone

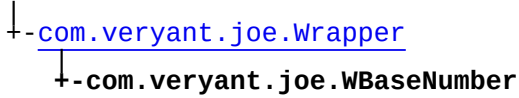
```
public java.lang.Object clone()
```

Returns a copy of this array

com.veryant.joe

Class WBaseNumber

java.lang.Object



All Implemented Interfaces:

[WNumber](#), InternalObject

Direct Known Subclasses:

[WLong](#), [WDouble](#), [WBigDecimal](#)

public abstract class **WBaseNumber**
 extends [Wrapper](#)
 implements InternalObject, [WNumber](#)

This class defines the methods that are common to all the numbers i.e int, short, long, double and BigDecimal.. Consider that in JOE some operators symbols are equivalent to methods names, i.e.

add	+
subtract	-
multiply	*
divide	/
remainder	%
equals	=
gt	>
lt	<
ge	>=
le	<=
ne	<>

Constructor Summary

public	WBaseNumber ()
--------	--------------------------------

Method Summary

java.math.BigDecimal	add (WNumber n)
java.math.BigDecimal	divide (WNumber n)
boolean	equals (java.lang.Object n)
WBoolean	equals (WNumber n)
WBoolean	ge (WNumber n)
WBoolean	gt (WNumber n)

WBoolean	le (WNumber n)
WBoolean	lt (WNumber n)
java.math.BigDecimal	multiply (WNumber n)
WBoolean	ne (WNumber n)
java.math.BigDecimal	remainder (WNumber n)
java.math.BigDecimal	subtract (WNumber n)
WString	toChar ()

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

WBaseNumber

```
public WBaseNumber()
```

Methods

add

```
public java.math.BigDecimal add(WNumber n)
```

subtract

```
public java.math.BigDecimal subtract(WNumber n)
```

multiply

```
public java.math.BigDecimal multiply(WNumber n)
```

(continued from last page)

divide

```
public java.math.BigDecimal divide(WNumber n)
```

remainder

```
public java.math.BigDecimal remainder(WNumber n)
```

equals

```
public boolean equals(java.lang.Object n)
```

equals

```
public WBoolean equals(WNumber n)
```

ne

```
public WBoolean ne(WNumber n)
```

lt

```
public WBoolean lt(WNumber n)
```

le

```
public WBoolean le(WNumber n)
```

gt

```
public WBoolean gt(WNumber n)
```

ge

```
public WBoolean ge(WNumber n)
```

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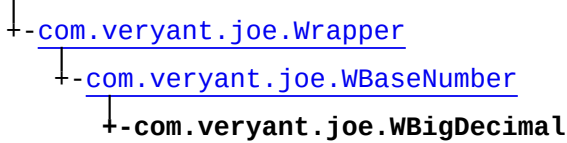
toChar

```
public WString toChar()
```

com.veryant.joe

Class WBigDecimal

java.lang.Object



All Implemented Interfaces:

[WNumber](#), [InternalObject](#), [WNumber](#)

public class **WBigDecimal**

extends [WBaseNumber](#)

implements [WNumber](#), [InternalObject](#), [WNumber](#)

Constructor Summary

public	WBigDecimal (java.math.BigDecimal n)
public	WBigDecimal (java.lang.Integer n)
public	WBigDecimal (java.lang.Double n)
public	WBigDecimal (java.lang.String n)

Method Summary

java.math.BigDecimal	abs ()
java.math.BigDecimal	bigDecimalValue ()
byte	byteValue ()
char	charValue ()
double	doubleValue ()
float	floatValue ()
java.lang.Object	getWrapped ()
int	intValue ()
long	longValue ()

java.math.BigDecimal	movePointLeft (WInteger newScale)
java.math.BigDecimal	movePointRight (WInteger newScale)
java.math.BigDecimal	negate ()
java.math.BigDecimal	pow (WInteger n)
int	precision ()
int	scale ()
java.math.BigDecimal	setScale (WInteger newScale)
java.math.BigDecimal	setScale (WInteger newScale, WInteger roundingMode)
short	shortValue ()
int	signum ()
java.math.BigDecimal	stripTrailingZeros ()
java.lang.String	toString ()
Wrapper.Type	type ()

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

(continued from last page)

WBigDecimal

```
public WBigDecimal(java.math.BigDecimal n)
```

WBigDecimal

```
public WBigDecimal(java.lang.Integer n)
```

WBigDecimal

```
public WBigDecimal(java.lang.Double n)
```

WBigDecimal

```
public WBigDecimal(java.lang.String n)
```

Methods

type

```
public Wrapper.Type type()
```

getWrapped

```
public java.lang.Object getWrapped()
```

bigDecimalValue

```
public java.math.BigDecimal bigDecimalValue()
```

byteValue

```
public byte byteValue()
```

charValue

```
public char charValue()
```

(continued from last page)

shortValue

```
public short shortValue()
```

intValue

```
public int intValue()
```

longValue

```
public long longValue()
```

floatValue

```
public float floatValue()
```

doubleValue

```
public double doubleValue()
```

toString

```
public java.lang.String toString()
```

abs

```
public java.math.BigDecimal abs()
```

negate

```
public java.math.BigDecimal negate()
```

pow

```
public java.math.BigDecimal pow(WInteger n)
```

precision

```
public int precision()
```

scale

```
public int scale()
```

movePointLeft

```
public java.math.BigDecimal movePointLeft(WInteger newScale)
```

movePointRight

```
public java.math.BigDecimal movePointRight(WInteger newScale)
```

setScale

```
public java.math.BigDecimal setScale(WInteger newScale)
```

setScale

```
public java.math.BigDecimal setScale(WInteger newScale,  
    WInteger roundingMode)
```

signum

```
public int signum()
```

stripTrailingZeros

```
public java.math.BigDecimal stripTrailingZeros()
```

com.veryant.joe

Class WBoolean

java.lang.Object

└ [com.veryant.joe.Wrapper](#)

└ [com.veryant.joe.WBoolean](#)

All Implemented Interfaces:
InternalObject

public class **WBoolean**
extends [Wrapper](#)

Field Summary

public static final	FALSE
public static final	TRUE

Method Summary

WBoolean	and (Block m)
WBoolean	and (WBoolean b)
boolean	booleanValue ()
boolean	equals (java.lang.Object b)
java.lang.Object	getWrapped ()
java.lang.Object	ifFalse (Block bTrue)
java.lang.Object	ifFalse (Block bTrue, Block bFalse)
java.lang.Object	ifTrue (Block bTrue)
java.lang.Object	ifTrue (Block bTrue, Block bFalse)
java.lang.Object	iif (java.lang.Object oTrue, java.lang.Object oFalse)
boolean	ne (WBoolean b)
WBoolean	not ()

WBoolean	or (Block m)
WBoolean	or (WBoolean b)
java.lang.String	toString ()
Wrapper.Type	type ()
WBoolean	xor (Block m)
WBoolean	xor (WBoolean b)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Fields

TRUE

public static final com.veryant.joe.WBoolean **TRUE**

FALSE

public static final com.veryant.joe.WBoolean **FALSE**

Methods

type

public Wrapper.Type **type**()

getWrapped

public java.lang.Object **getWrapped**()

booleanValue

public boolean **booleanValue**()

equals

```
public boolean equals(java.lang.Object b)
```

ne

```
public boolean ne(WBoolean b)
```

and

```
public WBoolean and(WBoolean b)
```

and

```
public WBoolean and(Block m)  
    throws JOEException
```

or

```
public WBoolean or(WBoolean b)
```

or

```
public WBoolean or(Block m)  
    throws JOEException
```

xor

```
public WBoolean xor(WBoolean b)
```

xor

```
public WBoolean xor(Block m)  
    throws JOEException
```

not

```
public WBoolean not()
```

iif

```
public java.lang.Object iif(java.lang.Object oTrue,  
                             java.lang.Object oFalse)  
    throws JOEException
```

ifTrue

```
public java.lang.Object ifTrue(Block bTrue)  
    throws JOEException
```

ifTrue

```
public java.lang.Object ifTrue(Block bTrue,  
                                Block bFalse)  
    throws JOEException
```

ifFalse

```
public java.lang.Object ifFalse(Block bTrue,  
                                 Block bFalse)  
    throws JOEException
```

ifFalse

```
public java.lang.Object ifFalse(Block bTrue)  
    throws JOEException
```

toString

```
public java.lang.String toString()
```

com.veryant.joe

Class WByteArray

java.lang.Object

└ [com.veryant.joe.Wrapper](#)

└ **com.veryant.joe.WByteArray**

All Implemented Interfaces:

InternalObject

public class **WByteArray**
extends [Wrapper](#)

This class implements a JOE array and it is a wrapper of a Java byte array.

Constructor Summary

public	WByteArray (byte[] array)
public	WByteArray (WNumber len)
public	WByteArray (int len)

Method Summary

java.lang.Object	add (WNumber obj) Returns a new array longer by one of this array with <i>obj</i> in the last position.
java.lang.Object	clone () Returns a copy of this array
java.lang.Object	get (WNumber idx) Returns the object at position <i>idx</i> .
byte[]	getWrapped () This is for internal use only
WInteger	length () Returns the length of this array.
java.lang.Object	set (int idx, WNumber obj) Sets the object <i>obj</i> at position <i>idx</i> .
java.lang.Object	set (WNumber idx, WNumber obj)
java.lang.Object	shift () Returns a new array shorter by one of this array without the object in the first position.
java.lang.Object	shift (int shft) Returns a new array shorter by <i>shft</i> of this array without <i>shft</i> objects at the beginning.

java.lang.Object	shift (WNumber shiftValue)
WInteger	size () Returns the length of this array.
java.lang.Object	slice (WNumber from, WNumber to) Returns the specified range of this array as a new array.
Wrapper.Type	type ()
java.lang.Object	unshift (WNumber obj) Returns a new array longer by one of this array with <i>obj</i> in the first position.

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Constructors

WByteArray

public **WByteArray**(byte[] array)

WByteArray

public **WByteArray**([WNumber](#) len)

WByteArray

public **WByteArray**(int len)

Methods

type

public Wrapper.Type **type**()

getWrapped

public byte[] **getWrapped**()

This is for internal use only

length

```
public WInteger length()
```

Returns the length of this array.

size

```
public WInteger size()
```

Returns the length of this array.

See Also:

[length\(\)](#)

get

```
public java.lang.Object get(WNumber idx)
```

Returns the object at position *idx*.

set

```
public java.lang.Object set(int idx,  
    WNumber obj)
```

Sets the object *obj* at position *idx*.

set

```
public java.lang.Object set(WNumber idx,  
    WNumber obj)
```

See Also:

[set\(int, WNumber\)](#)

add

```
public java.lang.Object add(WNumber obj)
```

Returns a new array longer by one of this array with *obj* in the last position.

shift

```
public java.lang.Object shift(int shft)
```

Returns a new array shorter by *shft* of this array without *shft* objects at the beginning.

shift

```
public java.lang.Object shift(WNumber shiftValue)
```

See Also:

[shift\(int\)](#)

shift

```
public java.lang.Object shift()
```

Returns a new array shorter by one of this array without the object in the first position.

unshift

```
public java.lang.Object unshift(WNumber obj)
```

Returns a new array longer by one of this array with *obj* in the first position.

slice

```
public java.lang.Object slice(WNumber from,  
                               WNumber to)
```

Returns the specified range of this array as a new array. The value at *from* is placed at the first position in the new array. The length of the returned array will be *to* - *from*.

clone

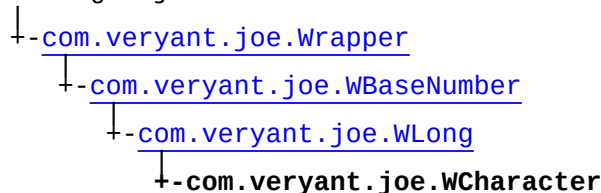
```
public java.lang.Object clone()
```

Returns a copy of this array

com.veryant.joe

Class WCharacter

java.lang.Object



All Implemented Interfaces:

InternalObject, [WNumber](#)

public class **WCharacter**

extends [WLong](#)

Constructor Summary

public	WCharacter (long n)
public	WCharacter (java.lang.String n)

Method Summary

java.lang.String	add (java.lang.Object s)
java.lang.String	add (WString s)
java.lang.Object	getWrapped ()
java.lang.String	toString ()
Wrapper.Type	type ()

Methods inherited from class [com.veryant.joe.WLong](#)

[abs](#), [add](#), [add](#), [add](#), [and](#), [and](#), [bigDecimalValue](#), [byteValue](#), [charValue](#), [divide](#), [divide](#), [divide](#), [doubleValue](#), [equals](#), [equals](#), [equals](#), [floatValue](#), [ge](#), [ge](#), [ge](#), [getWrapped](#), [gt](#), [gt](#), [gt](#), [intValue](#), [le](#), [le](#), [le](#), [longValue](#), [lt](#), [lt](#), [lt](#), [multiply](#), [multiply](#), [multiply](#), [ne](#), [ne](#), [ne](#), [negate](#), [not](#), [or](#), [or](#), [pow](#), [remainder](#), [remainder](#), [remainder](#), [shifta](#), [shiftr](#), [shiftr](#), [shortValue](#), [signum](#), [subtract](#), [subtract](#), [subtract](#), [toBinaryString](#), [toHexString](#), [toString](#), [type](#), [xor](#), [xor](#)

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class `java.lang.Object`

`equals`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

WCharacter

```
public WCharacter(long n)
```

WCharacter

```
public WCharacter(java.lang.String n)
```

Methods

type

```
public Wrapper.Type type()
```

getWrapped

```
public java.lang.Object getWrapped()
```

add

```
public java.lang.String add(WString s)
```

add

```
public java.lang.String add(java.lang.Object s)
```

toString

```
public java.lang.String toString()
```

(continued from last page)

com.veryant.joe Class WCharArray

java.lang.Object

└ [com.veryant.joe.Wrapper](#)

└ **com.veryant.joe.WCharArray**

All Implemented Interfaces:
InternalObject

public class **WCharArray**
extends [Wrapper](#)

This class implements a JOE array and it is a wrapper of a Java char array.

Constructor Summary

public	WCharArray (char[] array)
public	WCharArray (WNumber len)
public	WCharArray (int len)

Method Summary

java.lang.Object	add (WNumber obj) Returns a new array longer by one of this array with <i>obj</i> in the last position.
java.lang.Object	clone () Returns a copy of this array
java.lang.Object	get (WNumber idx) Returns the object at position <i>idx</i> .
char[]	getWrapped () This is for internal use only
WInteger	length () Returns the length of this array.
java.lang.Object	set (int idx, WNumber obj) Sets the object <i>obj</i> at position <i>idx</i> .
java.lang.Object	set (WNumber idx, WNumber obj)
java.lang.Object	shift () Returns a new array shorter by one of this array without the object in the first position.
java.lang.Object	shift (int shft) Returns a new array shorter by <i>shft</i> of this array without <i>shft</i> objects at the beginning.

java.lang.Object	shift (WNumber shiftValue)
WInteger	size () Returns the length of this array.
java.lang.Object	slice (WNumber from, WNumber to) Returns the specified range of this array as a new array.
java.lang.String	toString ()
Wrapper.Type	type ()
java.lang.Object	unshift (WNumber obj) Returns a new array longer by one of this array with <i>obj</i> in the first position.

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Constructors

WCharArray

```
public WCharArray(char[] array)
```

WCharArray

```
public WCharArray(WNumber len)
```

WCharArray

```
public WCharArray(int len)
```

Methods

type

```
public Wrapper.Type type()
```

getWrapped

```
public char[] getWrapped()
```

(continued from last page)

This is for internal use only

length

```
public WInteger length()
```

Returns the length of this array.

size

```
public WInteger size()
```

Returns the length of this array.

See Also:

[length\(\)](#)

get

```
public java.lang.Object get(WNumber idx)
```

Returns the object at position *idx*.

set

```
public java.lang.Object set(int idx,  
    WNumber obj)
```

Sets the object *obj* at position *idx*.

set

```
public java.lang.Object set(WNumber idx,  
    WNumber obj)
```

See Also:

[set\(int, WNumber\)](#)

add

```
public java.lang.Object add(WNumber obj)
```

Returns a new array longer by one of this array with *obj* in the last position.

shift

```
public java.lang.Object shift(int shft)
```

Returns a new array shorter by *shft* of this array without *shft* objects at the beginning.

shift

```
public java.lang.Object shift(WNumber shiftValue)
```

(continued from last page)

See Also:[shift\(int\)](#)

shift

```
public java.lang.Object shift()
```

Returns a new array shorter by one of this array without the object in the first position.

unshift

```
public java.lang.Object unshift(WNumber obj)
```

Returns a new array longer by one of this array with *obj* in the first position.

slice

```
public java.lang.Object slice(WNumber from,  
                               WNumber to)
```

Returns the specified range of this array as a new array. The value at *from* is placed at the first position in the new array. The length of the returned array will be *to* - *from*.

toString

```
public java.lang.String toString()
```

clone

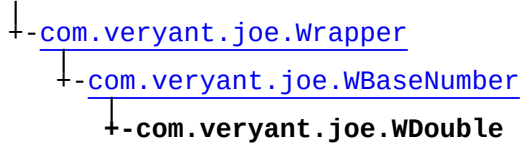
```
public java.lang.Object clone()
```

Returns a copy of this array

com.veryant.joe

Class WDouble

java.lang.Object



All Implemented Interfaces:

[WNumber](#), [InternalObject](#), [WNumber](#)

public class **WDouble**

extends [WBaseNumber](#)

implements [WNumber](#), [InternalObject](#), [WNumber](#)

Constructor Summary

public	WDouble (double n)
public	WDouble (java.lang.String n)

Method Summary

WDouble	abs()
WDouble	add (WDouble n)
WDouble	add (WInteger n)
WDouble	add (WLong n)
java.math.BigDecimal	bigDecimalValue()
byte	byteValue()
char	charValue()
WDouble	divide (WDouble n)
WDouble	divide (WInteger n)
WDouble	divide (WLong n)
double	doubleValue()

WBoolean	equals(WDouble n)
WBoolean	equals(WInteger n)
WBoolean	equals(WLong n)
float	floatValue()
WBoolean	ge(WDouble n)
WBoolean	ge(WInteger n)
WBoolean	ge(WLong n)
java.lang.Object	getWrapped()
WBoolean	gt(WDouble n)
WBoolean	gt(WInteger n)
WBoolean	gt(WLong n)
int	intValue()
WBoolean	le(WDouble n)
WBoolean	le(WInteger n)
WBoolean	le(WLong n)
long	longValue()
WBoolean	lt(WDouble n)
WBoolean	lt(WInteger n)
WBoolean	lt(WLong n)
WDouble	multiply(WDouble n)
WDouble	multiply(WInteger n)
WDouble	multiply(WLong n)
WBoolean	ne(WDouble n)
WBoolean	ne(WInteger n)

WBoolean	ne (WLong n)
WDouble	negate ()
WDouble	pow (WInteger e)
WDouble	remainder (WDouble n)
WDouble	remainder (WInteger n)
WDouble	remainder (WLong n)
short	shortValue ()
int	signum ()
WDouble	subtract (WDouble n)
WDouble	subtract (WInteger n)
WDouble	subtract (WLong n)
java.lang.String	toString ()
Wrapper.Type	type ()

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

(continued from last page)

WDouble

```
public WDouble(double n)
```

WDouble

```
public WDouble(java.lang.String n)
```

Methods

type

```
public Wrapper.Type type()
```

getWrapped

```
public java.lang.Object getWrapped()
```

byteValue

```
public byte byteValue()
```

charValue

```
public char charValue()
```

shortValue

```
public short shortValue()
```

intValue

```
public int intValue()
```

longValue

```
public long longValue()
```

(continued from last page)

floatValue

```
public float floatValue()
```

doubleValue

```
public double doubleValue()
```

bigDecimalValue

```
public java.math.BigDecimal bigDecimalValue()
```

signum

```
public int signum()
```

abs

```
public WDouble abs()
```

negate

```
public WDouble negate()
```

pow

```
public WDouble pow(WInteger e)
```

add

```
public WDouble add(WDouble n)
```

add

```
public WDouble add(WInteger n)
```

add

```
public WDouble add(WLong n)
```

(continued from last page)

subtract

```
public WDouble subtract(WDouble n)
```

subtract

```
public WDouble subtract(WInteger n)
```

subtract

```
public WDouble subtract(WLong n)
```

multiply

```
public WDouble multiply(WDouble n)
```

multiply

```
public WDouble multiply(WInteger n)
```

multiply

```
public WDouble multiply(WLong n)
```

divide

```
public WDouble divide(WDouble n)
```

divide

```
public WDouble divide(WInteger n)
```

divide

```
public WDouble divide(WLong n)
```

(continued from last page)

remainder

```
public WDouble remainder(WDouble n)
```

remainder

```
public WDouble remainder(WInteger n)
```

remainder

```
public WDouble remainder(WLong n)
```

equals

```
public WBoolean equals(WDouble n)
```

equals

```
public WBoolean equals(WInteger n)
```

equals

```
public WBoolean equals(WLong n)
```

lt

```
public WBoolean lt(WDouble n)
```

lt

```
public WBoolean lt(WInteger n)
```

lt

```
public WBoolean lt(WLong n)
```

gt

```
public WBoolean gt(WDouble n)
```

(continued from last page)

gtpublic [WBoolean](#) **gt**([WInteger](#) n)

gtpublic [WBoolean](#) **gt**([WLong](#) n)

gepublic [WBoolean](#) **ge**([WDouble](#) n)

gepublic [WBoolean](#) **ge**([WInteger](#) n)

gepublic [WBoolean](#) **ge**([WLong](#) n)

lepublic [WBoolean](#) **le**([WDouble](#) n)

lepublic [WBoolean](#) **le**([WInteger](#) n)

lepublic [WBoolean](#) **le**([WLong](#) n)

nepublic [WBoolean](#) **ne**([WDouble](#) n)

(continued from last page)

ne

```
public WBoolean ne(WInteger n)
```

ne

```
public WBoolean ne(WLong n)
```

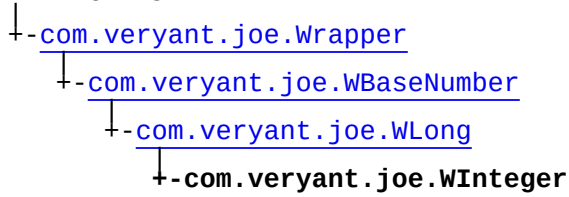
toString

```
public java.lang.String toString()
```

com.veryant.joe

Class WInteger

java.lang.Object



All Implemented Interfaces:

InternalObject, [WNumber](#)

public class **WInteger**

extends [WLong](#)

Constructor Summary

public	WInteger (long n)
public	WInteger (java.lang.String n)

Method Summary

WInteger	abs ()
WInteger	add (WInteger n)
WInteger	and (WInteger n)
WLong	and (WLong n)
WInteger	divide (WInteger n)
java.lang.Object	getWrapped ()
WInteger	multiply (WInteger n)
WInteger	negate ()
WInteger	not ()
WLong	or (WInteger n)
WLong	or (WLong n)

WInteger	pow(WInteger e)
WInteger	remainder(WInteger n)
WLong	shiftd(WInteger n)
WLong	shiftdl(WInteger n)
WLong	shiftr(WInteger n)
int	signum()
WInteger	subtract(WInteger n)
java.lang.String	toBinaryString()
java.lang.String	toHexString()
Wrapper.Type	type()
WLong	xor(WInteger n)
WLong	xor(WLong n)

Methods inherited from class [com.veryant.joe.WLong](#)

[abs](#), [add](#), [add](#), [add](#), [and](#), [and](#), [bigDecimalValue](#), [byteValue](#), [charValue](#), [divide](#), [divide](#), [divide](#), [doubleValue](#), [equals](#), [equals](#), [equals](#), [floatValue](#), [ge](#), [ge](#), [ge](#), [getWrapped](#), [gt](#), [gt](#), [gt](#), [intValue](#), [le](#), [le](#), [le](#), [longValue](#), [lt](#), [lt](#), [lt](#), [multiply](#), [multiply](#), [multiply](#), [ne](#), [ne](#), [ne](#), [negate](#), [not](#), [or](#), [or](#), [pow](#), [remainder](#), [remainder](#), [remainder](#), [shiftd](#), [shiftdl](#), [shiftr](#), [shortValue](#), [signum](#), [subtract](#), [subtract](#), [subtract](#), [toBinaryString](#), [toHexString](#), [toString](#), [type](#), [xor](#), [xor](#)

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

(continued from last page)

Constructors

WInteger

```
public WInteger(long n)
```

WInteger

```
public WInteger(java.lang.String n)
```

Methods

type

```
public Wrapper.Type type()
```

getWrapped

```
public java.lang.Object getWrapped()
```

add

```
public WInteger add(WInteger n)
```

subtract

```
public WInteger subtract(WInteger n)
```

multiply

```
public WInteger multiply(WInteger n)
```

divide

```
public WInteger divide(WInteger n)
```

remainder

```
public WInteger remainder(WInteger n)
```

and

```
public WLong and(WLong n)
```

and

```
public WInteger and(WInteger n)
```

or

```
public WLong or(WLong n)
```

or

```
public WLong or(WInteger n)
```

xor

```
public WLong xor(WLong n)
```

xor

```
public WLong xor(WInteger n)
```

not

```
public WInteger not()
```

shiftl

```
public WLong shiftl(WInteger n)
```

shiftr

```
public WLong shiftr(WInteger n)
```

shifta

```
public WLong shifta(WInteger n)
```

(continued from last page)

toHexString

```
public java.lang.String toHexString()
```

toBinaryString

```
public java.lang.String toBinaryString()
```

signum

```
public int signum()
```

abs

```
public WInteger abs()
```

negate

```
public WInteger negate()
```

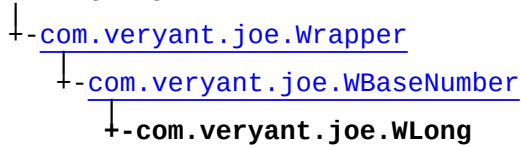
pow

```
public WInteger pow(WInteger e)
```

com.veryant.joe

Class WLong

java.lang.Object



All Implemented Interfaces:

InternalObject, [WNumber](#)

Direct Known Subclasses:

[WShort](#), [WInteger](#), [WCharacter](#)

public class **WLong**
 extends [WBaseNumber](#)

Constructor Summary

public	WLong (long n)
public	WLong (java.lang.String n)

Method Summary

WLong	abs ()
WDouble	add (WDouble n)
WLong	add (WInteger n)
WLong	add (WLong n)
WLong	and (WInteger n)
WLong	and (WLong n)
java.math.BigDecimal	bigDecimalValue ()
byte	byteValue ()
char	charValue ()
WDouble	divide (WDouble n)
WLong	divide (WInteger n)

WLong	divide(WLong n)
double	doubleValue()
WBoolean	equals(WDouble n)
WBoolean	equals(WInteger n)
WBoolean	equals(WLong n)
float	floatValue()
WBoolean	ge(WDouble n)
WBoolean	ge(WInteger n)
WBoolean	ge(WLong n)
java.lang.Object	getWrapped()
WBoolean	gt(WDouble n)
WBoolean	gt(WInteger n)
WBoolean	gt(WLong n)
int	intValue()
WBoolean	le(WDouble n)
WBoolean	le(WInteger n)
WBoolean	le(WLong n)
long	longValue()
WBoolean	lt(WDouble n)
WBoolean	lt(WInteger n)
WBoolean	lt(WLong n)
WDouble	multiply(WDouble n)
WLong	multiply(WInteger n)
WLong	multiply(WLong n)

WBoolean	ne (WDouble n)
WBoolean	ne (WInteger n)
WBoolean	ne (WLong n)
WLong	negate ()
WLong	not ()
WLong	or (WInteger n)
WLong	or (WLong n)
WLong	pow (WInteger e)
WDouble	remainder (WDouble n)
WLong	remainder (WInteger n)
WLong	remainder (WLong n)
WLong	shiftd (WInteger n)
WLong	shiftdl (WInteger n)
WLong	shiftr (WInteger n)
short	shortValue ()
int	signum ()
WDouble	subtract (WDouble n)
WLong	subtract (WInteger n)
WLong	subtract (WLong n)
java.lang.String	toBinaryString ()
java.lang.String	toHexString ()
java.lang.String	toString ()
Wrapper.Type	type ()
WLong	xor (WInteger n)

WLong	xor (WLong n)
-----------------------	--

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class `java.lang.Object`

`equals`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

WLong

`public WLong(long n)`

WLong

`public WLong(java.lang.String n)`

Methods

type

`public Wrapper.Type type()`

getWrapped

`public java.lang.Object getWrapped()`

byteValue

`public byte byteValue()`

(continued from last page)

charValue

```
public char charValue()
```

shortValue

```
public short shortValue()
```

intValue

```
public int intValue()
```

longValue

```
public long longValue()
```

floatValue

```
public float floatValue()
```

doubleValue

```
public double doubleValue()
```

bigDecimalValue

```
public java.math.BigDecimal bigDecimalValue()
```

signum

```
public int signum()
```

abs

```
public WLong abs()
```

negate

```
public WLong negate()
```

pow

```
public WLong pow(WInteger e)
```

add

```
public WLong add(WLong n)
```

add

```
public WLong add(WInteger n)
```

add

```
public WDouble add(WDouble n)
```

subtract

```
public WLong subtract(WLong n)
```

subtract

```
public WLong subtract(WInteger n)
```

subtract

```
public WDouble subtract(WDouble n)
```

multiply

```
public WLong multiply(WLong n)
```

multiply

```
public WLong multiply(WInteger n)
```

(continued from last page)

multiply

```
public WDouble multiply(WDouble n)
```

divide

```
public WLong divide(WLong n)
```

divide

```
public WLong divide(WInteger n)
```

divide

```
public WDouble divide(WDouble n)
```

remainder

```
public WLong remainder(WLong n)
```

remainder

```
public WLong remainder(WInteger n)
```

remainder

```
public WDouble remainder(WDouble n)
```

equals

```
public WBoolean equals(WLong n)
```

equals

```
public WBoolean equals(WInteger n)
```

equals

```
public WBoolean equals(WDouble n)
```

ltpublic [WBoolean](#) **lt**([WLong](#) n)

ltpublic [WBoolean](#) **lt**([WInteger](#) n)

ltpublic [WBoolean](#) **lt**([WDouble](#) n)

gtpublic [WBoolean](#) **gt**([WLong](#) n)

gtpublic [WBoolean](#) **gt**([WInteger](#) n)

gtpublic [WBoolean](#) **gt**([WDouble](#) n)

gepublic [WBoolean](#) **ge**([WLong](#) n)

gepublic [WBoolean](#) **ge**([WInteger](#) n)

gepublic [WBoolean](#) **ge**([WDouble](#) n)

(continued from last page)

le

```
public WBoolean le(WLong n)
```

le

```
public WBoolean le(WInteger n)
```

le

```
public WBoolean le(WDouble n)
```

ne

```
public WBoolean ne(WLong n)
```

ne

```
public WBoolean ne(WInteger n)
```

ne

```
public WBoolean ne(WDouble n)
```

and

```
public WLong and(WLong n)
```

and

```
public WLong and(WInteger n)
```

or

```
public WLong or(WLong n)
```

or

```
public WLong or(WInteger n)
```

xor

```
public WLong xor(WLong n)
```

xor

```
public WLong xor(WInteger n)
```

not

```
public WLong not()
```

shiftl

```
public WLong shiftl(WInteger n)
```

shiftr

```
public WLong shiftr(WInteger n)
```

shifta

```
public WLong shifta(WInteger n)
```

toString

```
public java.lang.String toString()
```

toHexString

```
public java.lang.String toHexString()
```

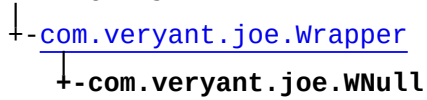
toBinaryString

```
public java.lang.String toBinaryString()
```

com.veryant.joe

Class WNull

java.lang.Object



All Implemented Interfaces:
InternalObject

public class **WNull**
extends [Wrapper](#)

Field Summary

public static final	value
---------------------	-----------------------

Method Summary

boolean	ge (java.lang.Object b)
java.lang.Object	getWrapped ()
boolean	gt (java.lang.Object b)
boolean	le (java.lang.Object b)
boolean	lt (java.lang.Object b)
boolean	ne (java.lang.Object b)
java.lang.String	toString ()
Wrapper.Type	type ()

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

[equals](#), [getClass](#), [hashCode](#), [notify](#), [notifyAll](#), [toString](#), [wait](#), [wait](#), [wait](#)

Fields

(continued from last page)

value

```
public static final com.veryant.joe.WNull value
```

Methods

ne

```
public boolean ne(java.lang.Object b)
```

gt

```
public boolean gt(java.lang.Object b)
```

ge

```
public boolean ge(java.lang.Object b)
```

lt

```
public boolean lt(java.lang.Object b)
```

le

```
public boolean le(java.lang.Object b)
```

getWrapped

```
public java.lang.Object getWrapped()
```

type

```
public Wrapper.Type type()
```

toString

```
public java.lang.String toString()
```

com.veryant.joe Interface WNumber

All Known Implementing Classes:

[WDouble](#), [WBigDecimal](#), [WBaseNumber](#)

public interface **WNumber**
extends

Method Summary

abstract java.math.BigDecimal	bigDecimalValue()
abstract byte	byteValue()
abstract char	charValue()
abstract double	doubleValue()
abstract float	floatValue()
abstract int	intValue()
abstract long	longValue()
abstract short	shortValue()
abstract int	signum()

Methods

byteValue

public abstract byte **byteValue()**

charValue

public abstract char **charValue()**

shortValue

public abstract short **shortValue()**

(continued from last page)

intValue

public abstract int **intValue()**

longValue

public abstract long **longValue()**

floatValue

public abstract float **floatValue()**

doubleValue

public abstract double **doubleValue()**

bigDecimalValue

public abstract java.math.BigDecimal **bigDecimalValue()**

signum

public abstract int **signum()**

com.veryant.joe Class Wrapper

java.lang.Object

└-com.veryant.joe.Wrapper

All Implemented Interfaces:

InternalObject

Direct Known Subclasses:

[WString](#), [WNull](#), [WCharArray](#), [WByteArray](#), [WBoolean](#), [WBaseNumber](#), [WArray](#)

public abstract class **Wrapper**

extends java.lang.Object

implements InternalObject

Constructor Summary

public	Wrapper()
--------	---------------------------

Method Summary

abstract java.lang.Object	getWrapped()
------------------------------	------------------------------

static Wrapper	newInstance (java.lang.Object obj)
--------------------------------	--

Methods inherited from class java.lang.Object

equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructors

Wrapper

public **Wrapper**()

Methods

newInstance

public static [Wrapper](#) **newInstance**(java.lang.Object obj)

(continued from last page)

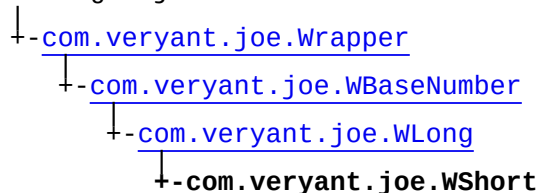
getWrapped

```
public abstract java.lang.Object getWrapped()
```

com.veryant.joe

Class WShort

java.lang.Object



All Implemented Interfaces:

InternalObject, [WNumber](#)

public class **WShort**

extends [WLong](#)

Constructor Summary

public	WShort (long n)
public	WShort (java.lang.String n)

Method Summary

java.lang.Object	getWrapped ()
java.lang.String	toHexString ()
Wrapper.Type	type ()

Methods inherited from class [com.veryant.joe.WLong](#)

[abs](#), [add](#), [add](#), [add](#), [and](#), [and](#), [bigDecimalValue](#), [byteValue](#), [charValue](#), [divide](#), [divide](#), [divide](#), [doubleValue](#), [equals](#), [equals](#), [equals](#), [floatValue](#), [ge](#), [ge](#), [ge](#), [getWrapped](#), [gt](#), [gt](#), [gt](#), [intValue](#), [le](#), [le](#), [le](#), [longValue](#), [lt](#), [lt](#), [lt](#), [multiply](#), [multiply](#), [multiply](#), [ne](#), [ne](#), [ne](#), [negate](#), [not](#), [or](#), [or](#), [pow](#), [remainder](#), [remainder](#), [remainder](#), [shifta](#), [shiffl](#), [shiftr](#), [shortValue](#), [signum](#), [subtract](#), [subtract](#), [subtract](#), [toBinaryString](#), [toHexString](#), [toString](#), [type](#), [xor](#), [xor](#)

Methods inherited from class [com.veryant.joe.WBaseNumber](#)

[add](#), [divide](#), [equals](#), [equals](#), [ge](#), [gt](#), [le](#), [lt](#), [multiply](#), [ne](#), [remainder](#), [subtract](#), [toChar](#)

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class java.lang.Object

`equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait`

Methods inherited from interface [com.veryant.joe.WNumber](#)

[bigDecimalValue](#), [byteValue](#), [charValue](#), [doubleValue](#), [floatValue](#), [intValue](#), [longValue](#), [shortValue](#), [signum](#)

Constructors

WShort

```
public WShort(long n)
```

WShort

```
public WShort(java.lang.String n)
```

Methods

type

```
public Wrapper.Type type()
```

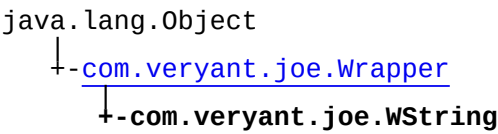
getWrapped

```
public java.lang.Object getWrapped()
```

toHexString

```
public java.lang.String toHexString()
```

com.veryant.joe
Class WString



All Implemented Interfaces:
InternalObject

public class **WString**
extends [Wrapper](#)

This class implements a JOE string. It implements most of the java.lang.String methods and has few more methods in order to simplify the comparison between strings, i.e. gt, ge, lt, le, ne, and other methods. Consider that in JOE some operators symbols are equivalent to methods names, i.e.

add	+
equals	=
gt	>
lt	<
ge	>=
le	<=
ne	<>

When a Java method outside the JOE interpreter is invoked, the instance of this object is converted in a java.lang.String.

Field Summary	
public final	value
Constructor Summary	
public	WString (java.lang.String s)
Method Summary	
WString	add (java.lang.Object obj) Returns a new string that concatenates the string representation of the specified object to the end of this string.
WString	add (WString str) Returns a new string that concatenates the specified string to the end of this string.
java.lang.String	at (WNumber index) Returns a new string containing the character at <i>index</i> position.
java.math.BigDecimal	bigDecimalValue () Returns a BigDecimal if this string contains a valid representation, null otherwise.
WCharacter	charAt (WNumber index)

WInteger	charCodeAt() Returns a integer with the codepoint of the first character of this string.
WInteger	charCodeAt(WNumber index) Returns a integer with the codepoint of the character at <i>index</i> position.
WInteger	compareTo(java.lang.String anotherString)
WInteger	compareTo(WString anotherString)
WInteger	compareToIgnoreCase(java.lang.String anotherString)
WInteger	compareToIgnoreCase(WString anotherString)
WString	concat(WString str)
WBoolean	contains(WString s)
java.lang.Double	doubleValue() Returns a double if this string contains a valid representation, null otherwise.
WBoolean	endsWith(WString s)
WBoolean	equals(WString s)
WBoolean	equalsIgnoreCase(WString s)
WBoolean	ge(WString s)
byte[]	getBytes()
byte[]	getBytes(java.nio.charset.Charset charset)
byte[]	getBytes(java.lang.String charsetName)
void	getChars(WInteger srcBegin, WInteger srcEnd, WCharArray dst, WInteger dstBegin)
java.lang.Object	getWrapped()
WBoolean	gt(WString s)
WInteger	indexOf(WString str)
WInteger	indexOf(WString str, WNumber fromIndex)
java.lang.Integer	intValue() Returns an integer if this string contains a valid representation, null otherwise.
WBoolean	isEmpty()

WInteger	lastIndexOf(WString str)
WInteger	lastIndexOf(WString str, WNumber fromIndex)
WBoolean	le(WString s)
WInteger	length()
java.lang.Long	longValue() Returns a long integer if this string contains a valid representation, null otherwise.
WBoolean	lt(WString s)
WBoolean	matches(WString regex)
WBoolean	ne(WString s)
WString	replace(WString target, WString replacement)
WString	replaceAll(WString regex, WString replacement)
WString	replaceFirst(WString regex, WString replacement)
java.lang.String[]	split(WString regex)
WBoolean	startsWith(WString s)
WBoolean	startsWith(WString s, WNumber toffset)
WString	substring(WNumber beginIndex)
WString	substring(WNumber beginIndex, WNumber endIndex)
char[]	toCharArray() Returns a char[] with the contents of this string
java.lang.String	toJava()
WString	toLowerCase()
java.lang.String	toString()
WString	toUpperCase()
WString	trim()
Wrapper.Type	type()

Methods inherited from class [com.veryant.joe.Wrapper](#)

[getWrapped](#), [newInstance](#)

Methods inherited from class `java.lang.Object`

`equals`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Fields

value

`public final java.lang.String value`

Constructors

WString

`public WString(java.lang.String s)`

Methods

type

`public Wrapper.Type type()`

getWrapped

`public java.lang.Object getWrapped()`

equals

`public WBoolean equals(WString s)`

lt

`public WBoolean lt(WString s)`

gt

`public WBoolean gt(WString s)`

(continued from last page)

ge

```
public WBoolean ge(WString s)
```

le

```
public WBoolean le(WString s)
```

ne

```
public WBoolean ne(WString s)
```

concat

```
public WString concat(WString str)
```

add

```
public WString add(WString str)
```

Returns a new string that concatenates the specified string to the end of this string.

add

```
public WString add(java.lang.Object obj)
```

Returns a new string that concatenates the string representation of the specified object to the end of this string.

startsWith

```
public WBoolean startsWith(WString s)
```

startsWith

```
public WBoolean startsWith(WString s,  
    WNumber toffset)
```

substring

```
public WString substring(WNumber beginIndex)
```

(continued from last page)

substring

```
public WString substring(WNumber beginIndex,  
                        WNumber endIndex)
```

charAt

```
public WCharacter charAt(WNumber index)
```

toLowerCase

```
public WString toLowerCase()
```

toUpperCase

```
public WString toUpperCase()
```

length

```
public WInteger length()
```

compareTo

```
public WInteger compareTo(java.lang.String anotherString)
```

compareToIgnoreCase

```
public WInteger compareToIgnoreCase(java.lang.String anotherString)
```

compareTo

```
public WInteger compareTo(WString anotherString)
```

compareToIgnoreCase

```
public WInteger compareToIgnoreCase(WString anotherString)
```

contains

```
public WBoolean contains(WString s)
```

endsWith

```
public WBoolean endsWith(WString s)
```

equalsIgnoreCase

```
public WBoolean equalsIgnoreCase(WString s)
```

indexOf

```
public WInteger indexOf(WString str)
```

indexOf

```
public WInteger indexOf(WString str,  
    WNumber fromIndex)
```

isEmpty

```
public WBoolean isEmpty()
```

lastIndexOf

```
public WInteger lastIndexOf(WString str)
```

lastIndexOf

```
public WInteger lastIndexOf(WString str,  
    WNumber fromIndex)
```

matches

```
public WBoolean matches(WString regex)
```

replace

```
public WString replace(WString target,  
    WString replacement)
```

replaceAll

```
public WString replaceAll(WString regex,  
    WString replacement)
```

replaceFirst

```
public WString replaceFirst(WString regex,  
    WString replacement)
```

split

```
public java.lang.String[] split(WString regex)
```

trim

```
public WString trim()
```

toJava

```
public java.lang.String toJava()
```

toString

```
public java.lang.String toString()
```

getBytes

```
public byte[] getBytes()
```

getBytes

```
public byte[] getBytes(java.nio.charset.Charset charset)
```

getBytes

```
public byte[] getBytes(java.lang.String charsetName)  
    throws java.lang.Exception
```

getChars

```
public void getChars(WInteger srcBegin,  
    WInteger srcEnd,  
    WCharArray dst,  
    WInteger dstBegin)
```

intValue

```
public java.lang.Integer intValue()
```

Returns an integer if this string contains a valid representation, null otherwise.

longValue

```
public java.lang.Long longValue()
```

Returns a long integer if this string contains a valid representation, null otherwise.

doubleValue

```
public java.lang.Double doubleValue()
```

Returns a double if this string contains a valid representation, null otherwise.

bigDecimalValue

```
public java.math.BigDecimal bigDecimalValue()
```

Returns a BigDecimal if this string contains a valid representation, null otherwise.

at

```
public java.lang.String at(WNumber index)
```

Returns a new string containing the character at *index* position.

charCodeAt

```
public WInteger charCodeAt(WNumber index)
```

Returns a integer with the codepoint of the character at *index* position.

charCodeAt

```
public WInteger charCodeAt()
```

Returns a integer with the codepoint of the first character of this string.

toCharArray

```
public char[] toCharArray()
```

Returns a char[] with the contents of this string

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